

Motion Control of a Hybrid Self-Reconfigurable Wheel-Legged Dual-Arm Robot

Rui Zhang, Hong Du, Peng Qiu, Yi Yang and Wenjie Song*

Abstract— Current wheeled bipedal robots face significant mobility challenges when traversing discontinuous terrain such as gaps and step-like obstacles, and suffer from substantial dynamic inefficiencies. This paper presents a hybrid self-reconfigurable wheel-legged dual-arm robot equipped with an active docking mechanism, enabling transitions between wheeled bipedal and multi-wheel-legged configurations. Based on a self-developed robotic platform, this work addresses key control challenges in articulated multi-wheel-legged mode and proposes a novel distributed operation paradigm for wheeled bipedal robots. Each module utilizes its manipulators for stable grasping of elevated objects and collaborative tasks, while the multi-unit system achieves efficient, high-load, and stable locomotion. To manage the control complexities in multimodal operation, we develop a unified modular control architecture integrating Virtual Model Control (VMC) and Linear Quadratic Regulator (LQR). For the articulated multi-wheel-legged mode, a body-posture controller regulates global body configuration, and a turning controller adjusts the wheelbase and roll angle via distributed actuation to manage the passive degrees of freedom (DoF) at the articulation points. Experimental validation using a physical prototype confirms the effectiveness and practicality of the proposed approach.

I. INTRODUCTION

Wheeled bipedal robots such as Skater [5], Nezha II [6], and BHR-WI [7] have evolved from two-link open-chain configurations, enabling enlarged workspaces and reduced leg inertia. Some robots integrate manipulators to extend the functional range and achieve arm-assisted balance [8]–[11]. The compliant wheeled quadrupedal robot Complios [12] adopts a continuous ground-contact strategy with load distribution and articulated waist steering. However, existing research has not yet combined wheeled bipedal robots with arm-based reconfiguration to develop an articulated multi-wheel-legged system. We propose a novel robot and a consistent modular control architecture for the reconfigurable system. Depending on task requirements and environmental conditions, each modular robot is capable of performing a range of manipulation tasks, while multi-unit systems can collaboratively reconfigure to traverse complex terrains with low energy consumption and improved stability.

The development of control theory for wheeled quadrupedal robots has lagged comparatively. Existing controller research primarily focuses on whole-body control

(WBC) [13]–[15] and model predictive control (MPC), while the application of LQR in wheeled quadrupedal robots remains relatively rare. To address this gap, we present a consistent modular control architecture applicable to both wheeled bipedal and multi-wheel-legged robots, with the goal of enhancing motion control performance and enabling efficient reconfiguration across different locomotion modes.

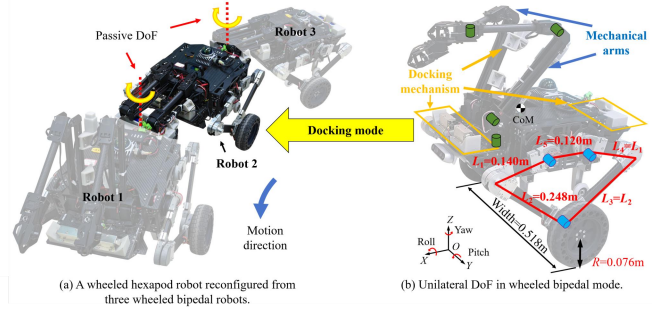


Fig. 1. Prototype of the hybrid self-reconfigurable wheel-legged dual-arm robot.

The main contributions of this paper can be summarized as follows:

- 1) A novel self-reconfigurable hybrid wheel-legged dual-arm robot is developed, featuring modular reconfiguration capability to transition into an articulated multi-wheel-legged mode. This design paradigm offers foundational insights for future distributed operation frameworks in bipedal robotic systems.
- 2) A consistent modular control framework is developed for both wheeled bipedal and multi-wheel-legged systems, with the operational feasibility rigorously validated through physical prototype experimentation.

II. MECHANISM DESCRIPTION

A. Overview

The hybrid self-reconfigurable wheel-legged dual-arm robot is a 14-DoF versatile modular platform designed for flexible operation in heavy-duty tasks. The overall architecture and module layout of a single unit are presented in Fig. 1(b). As a multifunctional platform, each unit supports wheel-legged walking, prone wheeled locomotion, and robotic arm-mediated docking with other modules to form an articulated multi-wheel-legged robotic system. The robot prototype comprises an alloy frame, five-link parallel wheel-legged mechanisms, dual robotic arms, and an active docking mechanism.

The front section integrates cooperative dual arms, which facilitate object retrieval in the prone posture and enable manipulation of elevated payloads while maintaining balanced

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standing, as shown in Fig. 2(a). Operated under position control, the robotic arms can transition to a docking interface mode, illustrated in Fig. 2(c). The rear section incorporates a docking mechanism that enables autonomous serial connection between units, forming a multi-articulated system (Fig. 2(d)). Equipped with depth cameras and LiDAR, each robot utilizes the docking mechanism to autonomously select and dock with other modules, thereby enhancing terrain adaptability and operational autonomy. The articulated multi-wheel-legged mode is designed for dynamic reconfiguration based on system states, enabling agile traversal over unstructured terrains such as gravel and grassland.

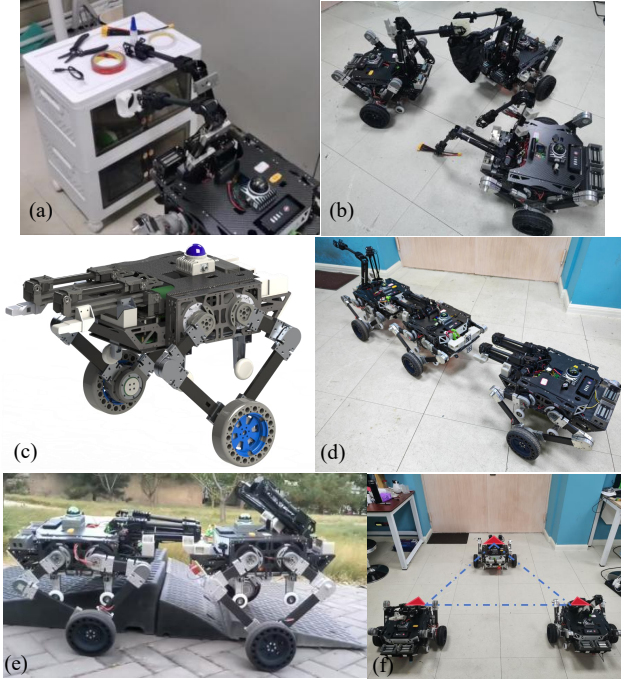


Fig. 2. Multimodal functionality of the robot. (a) Wheeled bipedal motion for grasping elevated objects. (b) Collaborative operation in prone wheeled mode. (c) Wheeled bipedal balanced docking mode. (d) Transition from the wheeled quadrupedal mode to the wheeled hexapodal mode. (e) Wheeled quadrupedal terrain adaption. (f) Triangular formation

The modular structural framework comprises three core modular strata: a main body integrated with standardized mechanical/electrical interfaces to enable rapid topological reconfiguration; five-link leg mechanisms for adapting ground clearance; and distributed sensing nodes that ensure real-time state synchronization across modules. Fig. 3 shows the architectural decomposition of the body assemblies.

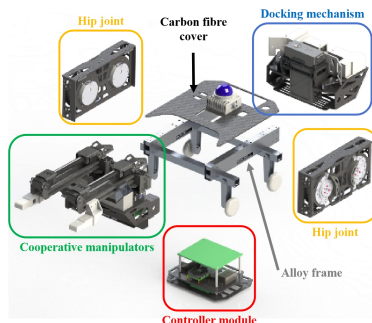


Fig. 3. Rendered image of body assemblies.

B. Docking Mechanism Design

The proposed docking mechanism, illustrated in Fig. 4, integrates wedge-profile active docking interfaces mounted on the robotic arms and complementary passive receptacles with bistable electromagnetic actuation, enabling dual-mode self-locking retention and controlled disengagement. This kinematic constraint system implements multi-axis motion constraints via geometrically interlocking prismatic guides, enforcing planar stability by eliminating redundant DoFs in both translational and rotational dimensions. The synergistic integration of mechanical interference fit and electromagnetic flux coupling ensures structural integrity under dynamic loading conditions.

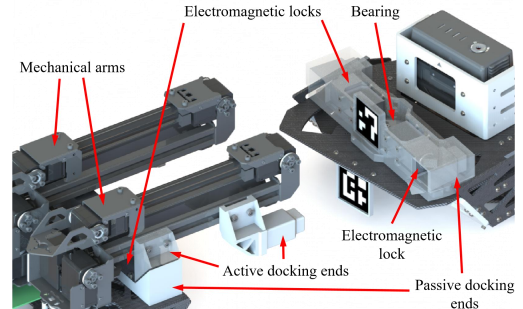


Fig. 4. Rendered image of the docking mechanism. The upper limit of articulation angle is constrained with a maximum value of 0.52 rad.

The reconfiguration interface incorporates a bimanual manipulation framework, utilizing 4-DoF manipulators with optimized kinematic chains to minimize collision risks while maintaining isostatic load distribution. As illustrated in Fig. 5, this architecture enables the formation of an articulated multi-wheel-legged morphology through coordinated arm-leg integration, where quasi-passive torsional compliance incorporated at the lumbar junction facilitates dynamic yaw modulation. Compared to conventional 8-DoF wheeled quadrupedal systems, this design achieves a minimum turning radius of 1.0 m via decoupled yaw articulations, while preserving structural stability through symmetric force-couple distribution across the docking mechanism.

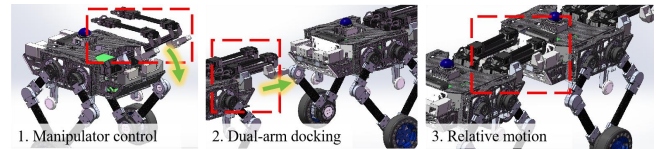


Fig. 5. Rendered image of the docking process.

C. Leg Design

To enhance operational versatility and mechanical robustness, the leg incorporates a five-link parallel mechanism actuated by co-axial drive modules, enabling full integration of hip motors within the body. This design eliminates transmission chains while preserving the dynamic workspace. Constructed with carbon fiber-reinforced tubes and aluminum alloy junctions, it achieves optimal strength-to-inertia ratios through material gradient optimization. Internally routed cables through hollow composite members enhance spatial efficiency, while duplex angular contact bearings at articulation points provide constrained axial load handling and parasitic motion rejection.

III. MODELING

A. Dynamic Modeling of Different Modes

In multi-wheel-legged mode, terrain adaptability is primarily governed by wheel contact distributions rather than pitch angle regulation. This mechanical transition decouples the sensitivity of body orientation from stability maintenance. We utilize a quasi-static analysis framework, discretizing the system into wheel-terrain interaction forces, virtual linkage reaction moments, and body inertial dynamics, as presented in Fig. 6. The end unit along the direction of motion is defined as $i = 1$. The force analysis of the body is detailed as follows:

$$F_{L,x,i} + F_{x,i-1 \rightarrow i} - F_{x,i \rightarrow i+1} = M \frac{\partial^2}{\partial t^2} [x + (L + L_M) \sin \theta] \quad (1)$$

$$F_{L,z,i} + F_{z,i-1 \rightarrow i} - F_{z,i \rightarrow i+1} - Mg = M \frac{\partial^2}{\partial t^2} [(L + L_M) \cos \theta] \quad (2)$$

where M is mass of each unit body. At the equilibrium point, θ and $F_{L,i}$ are approximated as equal. The state-space model (SSM) is derived through successive coordinate transformations and nonlinear term elimination. We present the LQR state vector as $\mathbf{X} = [\theta \ \dot{\theta} \ x \ \dot{x}]^T$, and the input vector as $\mathbf{U} = [\tau_w \ \tau_p]^T$.

The wheeled bipedal robot is modeled as a wheeled inverted pendulum (WIP). Kinematic consistency ensures identical Newton-Euler formulations for wheel-legged interactions across both multi-wheel-legged and wheeled bipedal modes. The force analysis of the body is as follows:

$$F_{L,x} = M \frac{\partial^2}{\partial t^2} [x + (L + L_M) \sin \theta - l \sin \varphi] \quad (3)$$

$$F_{L,z} - Mg = M \frac{\partial^2}{\partial t^2} [(L + L_M) \cos \theta + l \cos \varphi] \quad (4)$$

$$I_M \ddot{\varphi} = \tau_p + F_{L,x} l \cos \varphi + F_{L,z} l \sin \varphi \quad (5)$$

where I_M is the moment of inertia of each unit body. We present the LQR state vector for wheel-bipedal balance as $\mathbf{X}' = [\theta \ \dot{\theta} \ x \ \dot{x} \ \varphi \ \dot{\varphi}]^T$, and the input vector as $\mathbf{U}' = [\tau_w \ \tau_p]^T$.

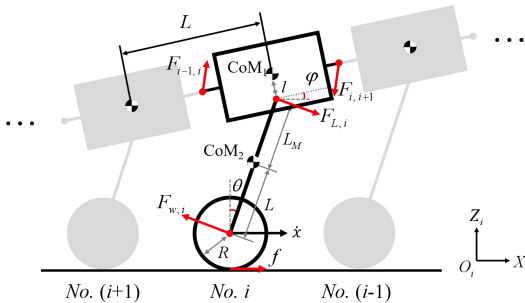


Fig. 6. The simplified model of a modular robot in multi-wheel-legged mode

B. Kinematic Modeling of Multi-Wheel-Legged System

The multi-wheel-legged robot prescribes kinematic DoF between adjacent axles through multi-axis revolute joints in the docking mechanism. Height adjustment is intentionally excluded from the dynamic formulation, with the system modeled as a quasi-rigid multibody system, and tire slip omitted, as illustrated in Fig. 7.

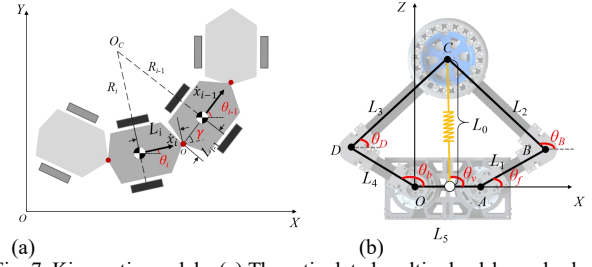


Fig. 7. Kinematic models. (a) The articulated multi-wheel-legged robot. (b) Five-link linkage parallel wheel-legged mechanisms.

Leveraging rigid-body kinematic constraints imposed by the articulated docking interface, adjacent robotic modules enforce velocity continuity at the shared articulation through the fundamental kinematic identity:

$$v_{i-1} = v_i \cos \gamma + \dot{\psi}_i L_r \sin \gamma \quad (6)$$

$$\dot{\psi}_{i-1} L_{i-1} = v_i \sin \gamma - \dot{\psi}_i L_r \cos \gamma \quad (7)$$

where v_{i-1} and v_i denote the longitudinal velocities of the front and rear robots respectively; with L_f and L_r representing the respective distances from the front and rear robot axles to the articulation; the yaw angles θ_{i-1} and θ_i and corresponding angular velocities $\dot{\theta}_{i-1}$ and $\dot{\theta}_i$ define the orientation dynamics of each module; γ is the articulation angle and is obtained as follows:

$$\gamma = \theta_{i-1} - \theta_i \quad (8)$$

C. Kinematic Modeling of five-link Linkage Parallel Leg

The coordinates of the foot-end, denoted as point C in Fig. 4(b), are analyzed and obtained as follows:

$$x_C = L_1 \cos \theta_b + L_2 \cos \theta_D \quad (9)$$

$$z_C = L_1 \sin \theta_b + L_2 \sin \theta_D \quad (10)$$

IV. MULTIMODAL MOTION CONTROL

A. Overall Control Framework

The control architecture integrates mode-specific SSMs with LQR-based feedback control to regulate both the hybrid state vector and leg length dynamics via VMC principles. Steering regulation, roll stabilization, and leg length modulation are achieved through PID control laws with adaptive gain scheduling. As shown in Fig. 8, this model-based framework ensures smooth motion transitions and reduces system complexity.

B. Consistent Modular Control Architecture

Due to the decoupling of balance control and leg control, the processes can be independently controlled. For the planar five-link mechanism with variable virtual-pole length, $\mathbf{K}$ is solved using a Riccati algebraic equation at each discrete interval and approximated using a seventh-order polynomial function as follows:

$$K_{ij}(L_0) = a_{ij} + b_{ij} L_0 + c_{ij} L_0^2 + \dots + h_{ij} L_0^7 \quad (11)$$

where $K_{ij}(L_0)$ represents the element in the i -th row and j -th column of the matrix $\mathbf{K}$. Thus, the control law is given by:

$$\mathbf{U} = \mathbf{K}(L_0)(\bar{\mathbf{X}} - \mathbf{X}) \quad (12)$$

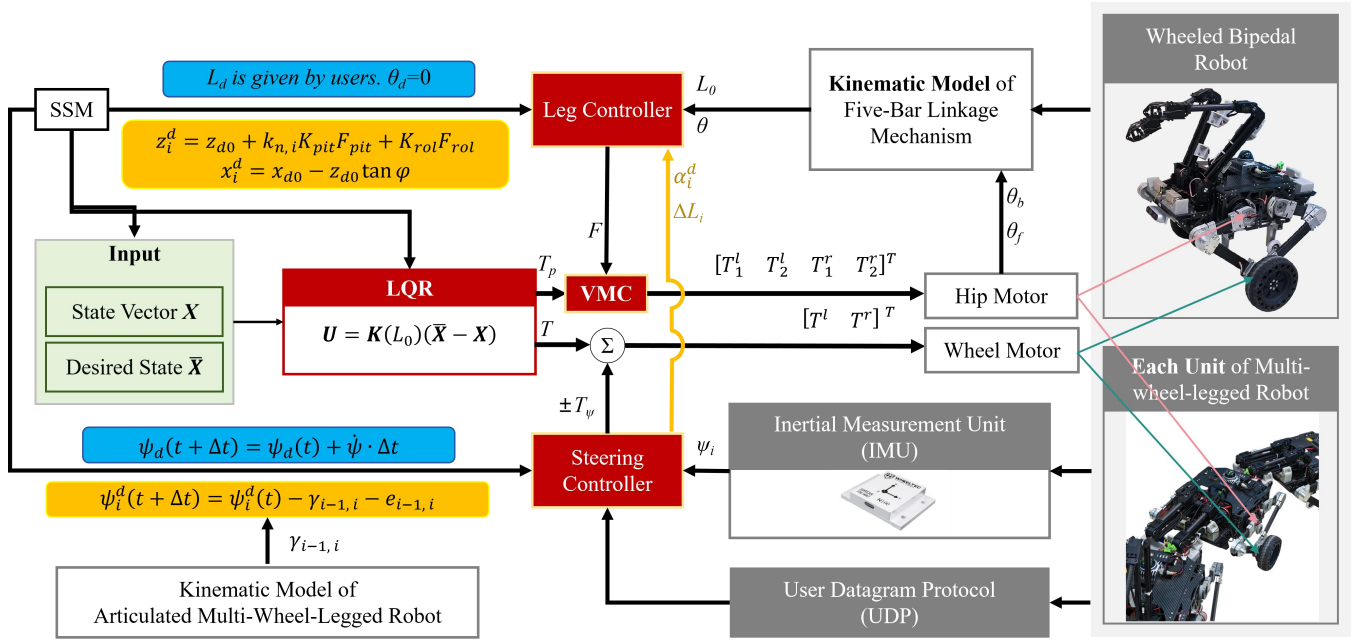


Fig. 8. Overview of the integrated control framework. The red, blue, and orange blocks denote controllers, desired variations in wheeled bipedal mode and multi-wheel-legged mode, respectively.

where the desired state in multi-wheel-legged mode is defined as $\bar{\mathbf{X}} = [\theta_d \ 0 \ x_d \ \dot{x}_d]^T$ and in wheeled bipedal mode as $\bar{\mathbf{X}}' = [0 \ 0 \ x_d \ \dot{x}_d \ 0 \ 0]^T$. It is necessary to control θ near the equilibrium point in the wheeled bipedal mode, as it is a WIP with inherent instability.

VMC enables precise tracking of the desired position and attitude. The workspace in the polar coordinates $\mathbf{x}_{pol} = [L_0 \ \theta_v]^T$ and joint space $\mathbf{q} = [\theta_b \ \theta_f]^T$ are geometrically related. Following the principle of VMC, we obtain:

$$\mathbf{T} \delta \mathbf{q} = \mathbf{F} \delta \mathbf{x} \quad (13)$$

Thus, the Jacobian matrix $\mathbf{J}$ of each module is identical and invariant in different modes and multi-wheel-legged modes as described as follows:

$$\delta \mathbf{x} = \mathbf{J} \delta \mathbf{q} \quad (14)$$

C. Body Posture Control in Multi-Wheel-Legged Mode

The workspace in the Cartesian coordinates is defined as $\mathbf{x}_{Cart} = [x \ z]^T$, which is intricately linked to the joint space $\mathbf{x}_{pol}$. The pitch angle and roll angle are independently controlled based on the positional feedback of each foot, with closed-loop control implemented for output tracking. Due to the serial kinematic chain, the output is determined by the following relationship:

$$z_i^d = z_{d0} \pm (n - 2i + 1)K_{pit}F_{pit} \mp K_{rol}F_{rol} \quad (15)$$

where K_{pit} and K_{rol} are the proportional coefficients respectively, and z_{d0} represents the original desired vertical height. During slope climbing, the front feet tend to lift and slip, so real-time feedback pitch angle φ is used to adjust the desired horizontal position x_i^d as shown in Fig. 9. The feedback pitch angle is redefined as:

$$\varphi_{ref} = \tan^{-1} \frac{z_{i-1} - z_i}{L} \quad (16)$$

where z_i is the feedback vertical height, and L is the distance between adjacent hinge axes.

D. Steering Control in Multi-Wheel-Legged Mode

The docking mechanism provides a passive DoF, forming an articulated multi-wheel-legged robot. During steering, the distributed drive system regulates the articulation angle and, leveraging the kinematic model, adjusts foot positioning to achieve the desired rotational velocity and turning radius.

The desired yaw angle ψ_i^d is obtained according to the recursive logic, as follows:

$$\psi_1^d(t + \Delta t) = \psi_1^d(t) + \dot{\psi}_d \cdot \Delta t \quad (17)$$

$$\psi_i^d(t + \Delta t) = \psi_{i-1}^d(t + \Delta t) - \gamma_{i-1,i} - e_{i-1,i} \quad (18)$$

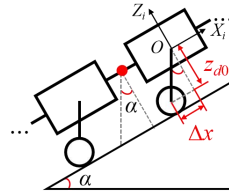


Fig. 9. Analysis of CoM adjustment during slope climbing.

where $e_{i-1,i}$ represents the original yaw angle offset. The yaw angle of each unit is controlled independently, with the output serving as compensation for the wheel torque to achieve distributed drive.

The longitudinal tire force is minimized with the inner wheelbase shortened and outer wheelbase extended. The yaw angle of each unit ψ_i is controlled by the PD controller with the wheelbase adjustment ΔL_i as the output to compensate x_i^d as follows:

$$x_i^d = x_{d0} - z_i \cdot \tan \varphi \pm \Delta L_i \quad (19)$$

where x_{d0} is the original desired horizontal position. The desired roll angle α_i^d is adjusted to facilitate turning, as given by:

$$\alpha_i^d = \tan^{-1} \left(\frac{\psi_d^2 R_i}{g} \right) \quad (20)$$

V. EXPERIMENTS

A. Prototype

The robot prototype and parameter table are shown in Fig. 10 and Table I below.

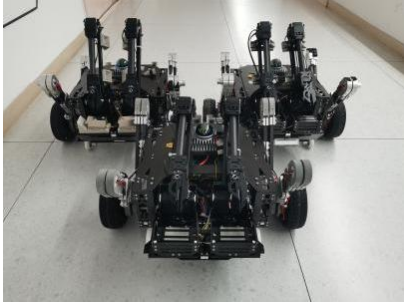


Fig. 10. The hybrid self-reconfigurable wheel-legged dual-arm robot in wheeled bipedal mode.

TABLE I: Technical Specifications of the robot

Category	Value
Total robot mass	11.65 kg
Width	0.5176 m
Length	0.5006 m
Maximum Standing Height	0.3815 m
Height when creeping	0.0891 m

B. Integrated Balance Control

The wheeled-bipedal motion capabilities are evaluated, including dynamic braking maneuvers and vertical height adjustments across three configurations (0.15 m, 0.175 m, and 0.20 m). As illustrated in Fig. 11(b), the state variables and corresponding leg length variations demonstrate stable transitions between operational modes. Under unloaded conditions, the robotic manipulator achieves object grasping with a maximum vertical reach of 0.6 m.

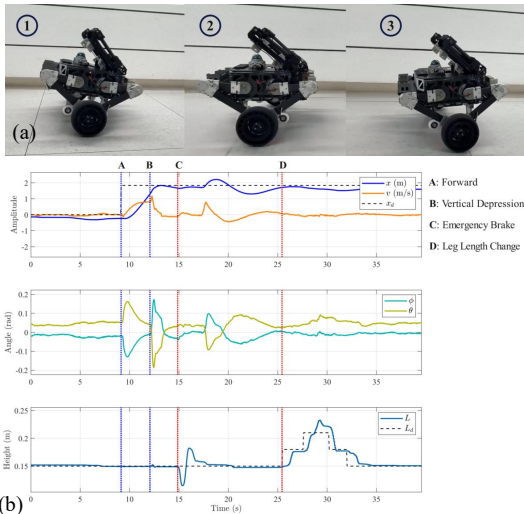


Fig. 11. Wheeled bipedal balance control. (a) Braking. (b) State variables and leg length curves.

Experiments of forward locomotion and vertical elevation control are performed on the hexapod/quadruped mode under standardized testing conditions. As depicted in Fig. 12(b), the state variables and centroidal height modulation profiles demonstrate stable configuration-dependent performance

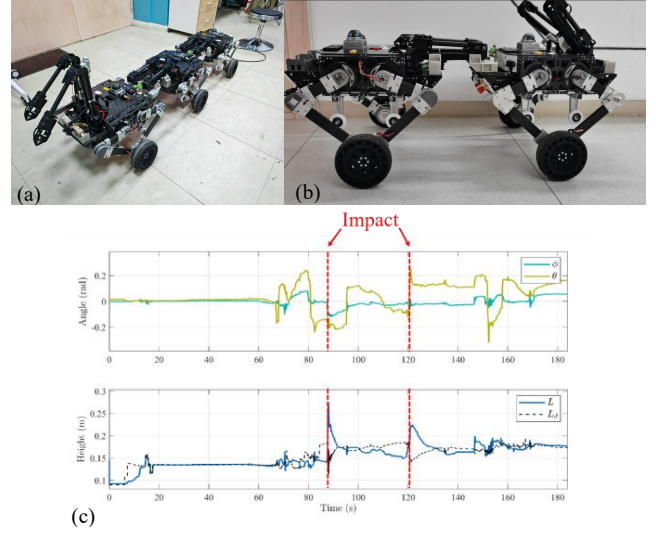
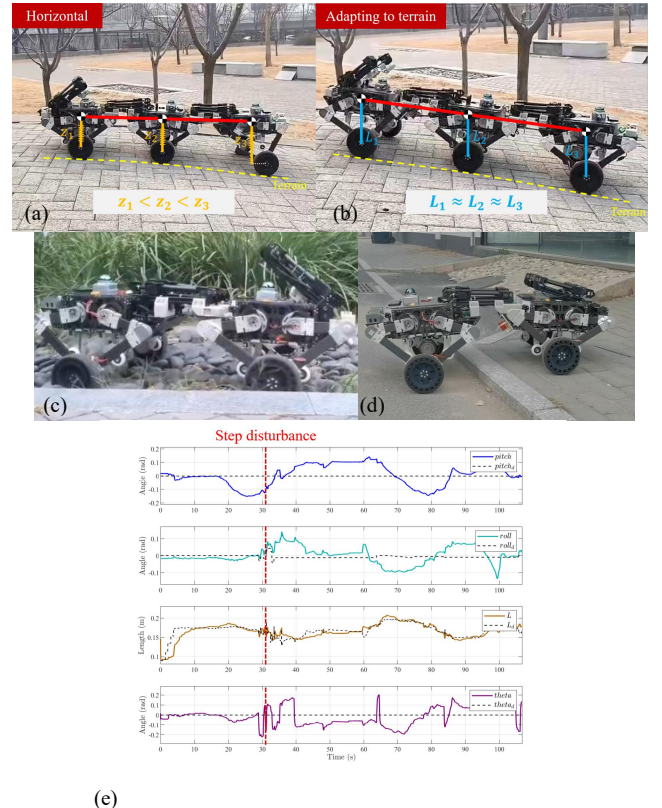


Fig. 12. Multi-wheel-legged balance control. (a) Wheeled hexapod balance. (b) Wheeled quadrupedal balance. (c) State variables and leg length curves before and after impact.

C. Body Posture Control in Multi-Wheel-Legged Mode

In the experiments, the wheeled hexapod robot maintains stability under two desired conditions: a horizontal body and adaptation to a 0.52-radian slope, with steady-state errors of 0.2 rad and 0.1 rad observed, respectively. Both angles recover to the desired range within 1 second after step disturbance.



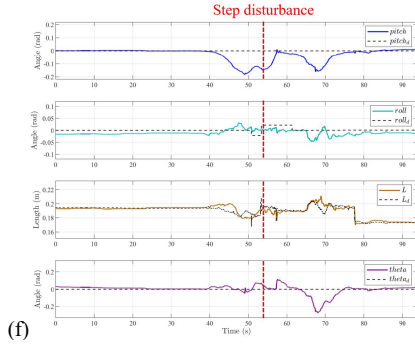


Fig. 13. Slope climbing. (a) Body horizontal. (b) Adapting to terrain. (c) 10 cm step crossing. (d) Gravel road traversal. (e) Curves at the horizontal state. (f) Curves during terrain adaptation.

D. Steering in the multi-wheel-legged mode

Fig. 14 presents experimental results of steering control performance, demonstrating yaw angle and angular velocity profiles during robotic maneuvers at turning radii of 1.0 m under a constant angular velocity of 1 rad/s. The experimental data indicate that both the yaw angular velocity and yaw angle achieve convergence to their target values within 0.5 seconds and 1 second, respectively.

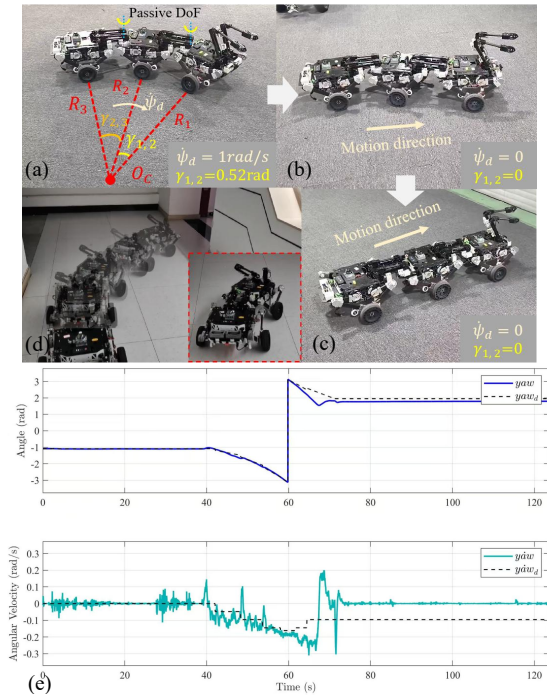


Fig. 14. Steering. (a) Wheeled hexapod steering. (b) Self-centering. (c) Move forward. (d) Wheeled quadrupedal steering. (e) Yaw angle and yaw angle velocity curves.

VI. CONCLUSION

This paper introduces a hybrid self-reconfigurable robotic architecture integrating wheel-legged mobility and dual-arm manipulation, synthesizing articulated multi-wheel-legged and wheeled bipedal modes through compliance-preserving active docking mechanisms. We formulate unified dynamic and kinematic equations, explicitly addressing nonholonomic constraints induced by the docking interface's passive DoF.

The consistent modular control architecture is proposed to reduce the complexity of reconfiguration. Body-posture and steering controllers for multi-wheel-legged configuration are designed to enhance stability and minimize energy consumption. The feasibility of the algorithm is validated through real-world experiments, confirming the effectiveness of the control framework.

In future work, we aim to apply reinforcement learning methods to practical applications and implement MPC or WBC to improve the precision and reliability of hybrid wheel-legged control.

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